🔑 **Answer Key — Teacher Copy · Lesson 4-1**

MSTAR PRACTICE WORKSHEET

Lesson 4-1: Understand Percent

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What does the word percent mean?

A. Per ten

B. Per thousand

C. Per hundred

D. Per part

Percent means per hundred, which is why every percent can be written as a fraction with a denominator of 100.

Question 2 SELECTED RESPONSE — CORRECT: D

3 out of every 10 students walk to school. What percent is that?

A. 3%

B. 10%

C. 13%

D. 30%

Multiply both terms of the ratio by 10 to compare to 100: 3 out of 10 becomes 30 out of 100, or 30%.

Question 3 SELECTED RESPONSE — CORRECT: C

A decimal grid has 100 squares and 65 are shaded. What percent is shaded?

A 6.5%

B 35%

C 65%

D 100%

Each square is one percent, so 65 shaded squares is 65%.

Question 4 SELECTED RESPONSE — CORRECT: D

Which percent means MORE than one whole?

A 0.8%

B 80%

C 100%

D 120%

100% is exactly one whole, so any percent above 100 is more than one whole.

Question 5 SELECTED RESPONSE — CORRECT: D

The orange kitten weighs 1.5 times as much as the gray kitten. What percent of the gray kitten's weight is that?

A 1.5%

B 15%

C 50%

D 150%

The gray kitten is the whole, or 100%. One and a half wholes is 100% + 50% = 150%.

Question 6 SELECTED RESPONSE — CORRECT: B

What percent is 9 out of 100?

A. 0.9%

B. 9%

C. 19%

D. 90%

A percent already compares to 100, so 9 out of 100 is exactly 9%.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A robotics club had a fundraising goal of $500. By Friday, they collected $750. What percent of their goal did they achieve?

A. 50%

B. 75%

C. 125%

D. 150%

$750 is 1.5 times the whole goal of $500. 1.5 × 100 = 150%, which represents 150% of the goal.

- **A:** That is the amount collected BEYOND the goal, not the percent of the goal reached.
- **B:** That compares $750 to $1000. The goal is $500, and $750 is more than the whole goal.
- **C:** That would be $625 out of $500. Divide the amount collected by the goal: 750 divided by 500.

Part B — correct answer: D

Which explanation best supports your answer to Part A?

A. Subtracting $500 from $750 gives $250, and 250 is 75% of 500.

B. Since $500 is half of $1,000, the percent must be 50%.

C. The ratio 500 to 750 simplifies to 2 to 3, which is 125%.

D. Dividing $750 by $500 gives 1.5, and multiplying 1.5 by 100 gives 150%, representing 1.5 times the original whole.

A percent compares a part to a whole of 100. 750 / 500 = 1.5 = 150/100 = 150%.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that correctly represent a rate per 100 or compare percents to one whole:

- **A** 100% represents exactly one whole quantity.
- **B** 45 out of 100 squares shaded on a 10×10 grid represents 45%.
- **C** A percent greater than 100% represents an amount that is more than the original whole.
- **D** 0.5% is equal to 50 out of 100.
- **E** A percent is a ratio that compares a number to 100.

Statements A, B, C, and E are true. Statement D is false because 0.5% is 0.5 out of 100 (1 out of 200), whereas 50 out of 100 is 50%.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A car dealership sold 50 cars last month: 20 sedans, 15 SUVs, 10 trucks, and 5 vans. The owner will order 100 cars for next month and wants the order to match last month's sales pattern. Write each type as a percent, then say how many of each to order — and explain why percents make this easier than the raw counts.

Model answer: Each count is out of 50, so multiply both terms by 2 to compare to 100: sedans 20/50 = 40/100 = 40%, SUVs 15/50 = 30/100 = 30%, trucks 10/50 = 20/100 = 20%, vans 5/50 = 10/100 = 10%. The percents total 100%, which confirms the whole is accounted for. Because the new order is exactly 100 cars, each percent IS the number to order: 40 sedans, 30 SUVs, 20 trucks, and 10 vans. Percents make this easier because they restate every count against the same whole of 100, so two months with different totals can be compared directly.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.